Baloyi Thabang Bongani Junior

Quantitative Risk Analytics | *Mathematical Statistics* | *Data & Model Controls*

079 346 8463 | Baloyibongani1@gmail.com

linkedin.com/in/thabang-bongani-junior-baloyi | github.com/thabangTheActuaryCoder

PROFESSIONAL SUMMARY

Quantitative professional with an MSc in Mathematical Statistics completed with distinction, an Honours degree in Statistics and Data Science with a minor in Actuarial Science, and CFA Level I in progress. More than five years of experience across statistical modelling, Monte Carlo simulation, model monitoring, data engineering, and production analytics, with research in derivative hedging under incomplete markets.

Strong in Python, SQL, VBA, Advanced Excel, and Power BI. Builds automated data pipelines, data-quality controls, and auditable decision workflows, investigates anomalies to root cause, and explains technical results clearly to non-technical stakeholders. Brings this foundation to counterparty credit risk, where accurate exposure data, explainable models, and disciplined controls support sound credit decisions.

EDUCATION

University of the Free State — **Jan 2026 to Oct 2026**
Master of Science, Mathematical Statistics — *80% (With Distinction)* | *Academic Record*

University of Cape Town — **Dec 2025**
BSc Honours, Statistics and Data Science, minor in Actuarial Science — *74%* | *Transcript* | *Proof of Qualification*

University of Cape Town — **Dec 2022**
Bachelor of Commerce, Management Studies — *70%* | *Transcript* | *Certificate*

University of the Free State — **2016 to 2017**
Bachelor of Science, Actuarial Sciences (Incomplete) — *Not completed* | *Academic Record*

RESEARCH

- **Mean-Variance Hedging with Neural Networks using VAR Sample Paths** (BSc Honours thesis, UCT, 2024, co-authored): hedged a European spread option on two correlated equities (Chevron, ExxonMobil) in an incomplete market. Fitted a VAR(11) model to log returns, simulated price paths, and trained a neural hedging strategy to minimise expected squared replication error. Benchmarked P&L against Black-Scholes delta hedging on 5,000 simulated paths and on market data: the network earned a higher mean P&L, while delta hedging delivered the lower P&L variance. Repository
- **Understanding Health Insurance Cost Predictions Using Deep Neural Networks, Monte Carlo Simulation and Shapley Values** (MSc dissertation, UFS, 2026): compared a deep neural network with a traditional pricing model and tree ensembles on held-out data, used Monte Carlo simulation to generate a full predicted cost distribution with validation diagnostics, and applied threshold-conditional Shapley attribution to show that upper-tail cost drivers differ from average drivers. Dissertation

TECHNICAL SKILLS

Quantitative & Risk Analytics: Monte Carlo Simulation, Derivative Hedging (Black-Scholes Delta, Spread Options), Mean-Variance Hedging, Tail Attribution, Statistical Modelling, Model Monitoring, Root Cause Analysis, Time Series Analysis (VAR), Regression, Multivariate and Bayesian Analysis, Portfolio Theory, Linear and Mixed-Integer Programming, Machine Learning

Excel, Reporting & BI: Advanced Microsoft Excel, VBA, Power BI, Tableau, KPI Reporting, Data Storytelling, Automated Dashboards

Programming: Python, SQL, R, MATLAB, STATA, C#, Java, NumPy, pandas, scikit-learn, TensorFlow, Keras, matplotlib, pytest, LaTeX

Data Engineering: ETL Pipelines, Data Validation, Data Quality Controls, PostgreSQL, Snowflake, Amazon Redshift, Spark, PySpark, Databricks, Delta Lake, dbt, Azure Data Factory, Microsoft Fabric

Cloud & Delivery: Azure, AWS, Docker, Git, GitHub Actions, Azure DevOps, Bitbucket Pipelines, SonarQube, Jira, Postman, WebdriverIO, REST APIs

WORK EXPERIENCE

Senior Full Stack AI Engineer
4S4, Cape Town, ZA (Full Time, Hybrid) *Jun 2026 to Present*

- **Clients:** Rheinmetall Denel Munition, Old Mutual Bank, Ooba, Deloitte
- Led development of EAPP, an enterprise planning platform integrating SAP-derived demand, stock and open supply, multi-level BOM explosion, finite work-centre capacity, and cross-plant scheduling (Live Demo)
- Built explainable decision workflows using Monte Carlo lead-time simulation, critical-path analysis, linear programming, and lexicographic mixed-integer programming to identify capacity constraints and recommend feasible production, overtime, transfer, and delivery plans
- Develop and maintain data integration pipelines, including SAP S/4HANA 2023-to-warehouse extraction, ETL workflows, and semantic layers
- Implemented a governed planning co-pilot with registered commands, structured outputs, audit trails, and human confirmation before actions are applied
- Translate client requirements into technical solutions and own features across architecture, implementation, testing, deployment, and support

Artificial Intelligence Engineer
Full Stack, Cape Town, ZA (Full Time) *Mar 2023 to May 2026*

- **Clients:** Virgin Active UK, Virgin Active SA, Expandly
- Built data and ML workflows using Python, SQL, Spark, PySpark, Databricks, and PostgreSQL, covering data modelling, batch processing, and analytics-ready outputs
- Integrated ML models into production services on AWS and Azure with monitoring, validation checks, and data quality controls
- Implemented testing, build, and code-quality controls using pytest, SonarQube, endpoint smoke tests, Bitbucket Pipelines, GitHub Actions, and Azure DevOps
- Built safety and compliance components, including guardrails, risk-pattern detection, emergency response routing, and auditable decision workflows

AI Machine Learning Engineer Intern
Full Stack, Cape Town, ZA *Jan 2023 to Mar 2023*

- Built and evaluated classification and regression models, including Logistic Regression, Random Forests, and XGBoost
- Reported model performance and data patterns using matplotlib and Power BI, and ran end-to-end automation tests with WebdriverIO covering data validation

Junior Data Scientist
Discovery Limited, Cape Town, ZA (Full Time) *Jan 2022 to Dec 2022*

- Established model monitoring protocols to track drift, performance degradation, and data pipeline integrity across production database environments.
- Developed, trained, and evaluated machine learning models using Python and SQL on large-scale healthcare and operational datasets to optimise client scheme services.
- Collaborated with clinical, actuarial, and business stakeholders to translate domain requirements into statistical solutions and communicated quantitative insights to non-technical users.

Data Analyst Intern
Discovery Limited, Cape Town, ZA *Jan 2021 to Dec 2021*

- Analysed large customer datasets using Advanced Excel and SQL to extract behavioural patterns and deliver actionable insights for customer engagement strategies.
- Collaborated cross-functionally with Health Actuarial, Business Intelligence, and Management Information (MI) teams to align quantitative findings with business objectives.
- Designed automated dashboards and recurring analytical reports to present core customer and health engagement metrics to stakeholders.

PROFESSIONAL QUALIFICATIONS

- CFA Level I (In Progress); Microsoft Certified: Fabric Data Engineer Associate (DP-700), 2026
- Actuarial Society of South Africa exemptions: A112 (Business Economics), A113 (Business Finance)

HONOURS & AWARDS

- Best Aggregate Average Actuarial Science and Dean's List, UFS; Golden Key; Moshal and Jurgen Klauss Bathe Scholarships